

LAKSHMAN PAVAN KUMAR BODAPATI

Software Engineer | Distributed Systems & Cloud Architect

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PROFESSIONAL SUMMARY

Software Engineer with 6+ years designing, building, and scaling high-performance distributed systems and cloud-native microservices for healthcare and enterprise environments. Expertise in cloud infrastructure optimization (AWS, Azure), cost reduction, and improving system reliability through automated DevOps pipelines and performance engineering.

KEY ACHIEVEMENTS

- Reduced system deployment downtime by 75% through microservices migration, serving 250K+ daily clinical trial requests at Medpace.
- Discovered and escalated a critical SQL Server bug (affecting versions 2019 and 2022) to Microsoft through systematic reproduction and production data analysis, resulting in official acknowledgment and patch release.
- Architected event-driven microservices processing 250K+ daily requests with <200ms p95 latency, implementing CQRS and saga patterns for distributed transaction management.
- Improved API response times by 40% using Redis caching and MySQL optimization for high-volume laboratory operations.
- Reduced Azure cloud storage costs by 30% through serverless automation, processing over 10,000 daily file operations.
- Enhanced HR data pipeline efficiency by 60% at ADP, supporting 500K+ global employees across 220 countries.
- Achieved 99.9% system uptime through comprehensive observability implementation with Datadog and ELK Stack.

TECHNICAL SKILLS:

- **Languages:** Java 8/17, Python, JavaScript (ES6+), TypeScript, SQL, HTML5/CSS
- **Java Technologies:** Spring Boot, Spring Cloud, Spring MVC, Spring Security, Spring Data JPA, Spring IOC, Hibernate
- **Frontend Technologies:** Angular 17, React, RxJS, NgRx, Bootstrap, JSON/XML, Responsive Design
- **Cloud Platforms:** AWS (EC2, S3, Lambda, RDS, CloudWatch, SNS, Redshift, Auto Scaling, Load Balancer), Azure (AKS, ACI, Functions, KeyVault, App Configuration, DevOps, Blob Storage)
- **Databases:** Oracle, MySQL, Redis, Amazon DynamoDB, SQL Server, MongoDB, Query Optimization, Indexing
- **DevOps & CI/CD:** Docker, Kubernetes, Jenkins, Azure DevOps, GitHub Actions, Helm Charts, YAML, Git, Bitbucket
- **Data Integration:** Apache Kafka, Apache Airflow, Talend ETL, RESTful APIs, GraphQL, Microservices Architecture
- **Monitoring & Testing:** Datadog, AWS CloudWatch, ELK Stack, JUnit 5, Mockito, Testcontainers, Cypress, Jasmine
- **Development Tools:** IntelliJ IDEA, Visual Studio/Code, Postman, Swagger/OpenAPI, SonarQube, ESLint, Confluence
- **Methodologies:** Agile (Scrum), Test-Driven Development (TDD), SDLC, Project Management (Jira, Azure DevOps)

PROFESSIONAL EXPERIENCE

Software Engineer II

Mar 2023 - Present

Medpace (Leading Global Clinical Research Organization)

Cincinnati, OH

- Led technical design reviews with stakeholders across Platform, QA, and Product teams, influencing architectural decisions for laboratory data modernization initiative.
- Architected migration from legacy monolithic Java application to 4 Spring Boot microservices deployed on Kubernetes (AKS), reducing deployment downtime by 75% and enabling independent service scaling.
- Built an enterprise API Gateway using Spring Cloud Gateway with Resilience4j circuit breaker patterns, reducing system latency by 15% and improving fault tolerance across distributed services.
- Designed and implemented Azure serverless solution using Azure Functions for automated blob storage lifecycle management, reducing cloud costs by 30% while processing 10K+ daily file cleanup operations.
- Developed 25+ high-performance RESTful APIs with Redis caching layer and MySQL query optimization, achieving 40% faster response times for sample processing workflows.
- Engineered automated data pipelines using Apache Kafka (event streaming), Apache Airflow (orchestration), and Talend ETL, reducing batch processing time by 40% and enabling real-time analytics for 5,000+ daily transactions.
- Resolved critical Kafka consumer lag and data loss issues through consumer group optimization and dead letter queue implementation, achieving 99.9% data integrity across streaming pipelines.
- Configured Helm charts for declarative Kubernetes deployments with environment-specific configurations, reducing rollback incidents by 15% across dev/staging/production environments.

- Designed comprehensive Datadog monitoring dashboards for microservices health metrics, API latency tracking (p50/p95/p99), and error rate monitoring, improving system uptime to 99.9%.
- Automated enterprise desktop application deployment using Azure DevOps release pipelines, distributing MSI packages via Azure File Share with versioning for 500+ end users.
- Designed and implemented a distributed caching strategy using Redis Cluster, reducing database load by 60% and improving API response times from 800ms to 320ms at p95.
- Implemented HIPAA-compliant data encryption and access control mechanisms for PHI (Protected Health Information) handling across microservices, ensuring regulatory compliance for clinical trial data processing.
- Optimized Angular 17 application using standalone components and OnPush change detection strategy, improving rendering speed by 30% and reducing page load time from 4.0s to 2.8s.
- Implemented end-to-end integration testing using JUnit 5, Testcontainers, and H2 in-memory database, ensuring reliable and isolated test execution in CI/CD pipelines.
- Mentored 3 junior engineers on microservices architecture patterns, Spring Boot best practices, and Kubernetes deployments, improving team code quality and reducing onboarding time by 40%.
- Participated in on-call rotation, diagnosing and resolving production incidents with an average MTTR of <30 minutes, maintaining 99.9% SLA.

Application Developer Intern

Jun 2022 - Aug 2022

ADP (Automatic Data Processing - HRM Platform)

Pasadena, CA

- Developed AWS Lambda functions for automated API load data generation and workflow optimization, enabling scalable performance testing.
- Implemented AWS S3 and Amazon Redshift data solutions, enhancing data accessibility and improving query performance for analytics workloads.
- Optimized Redshift query performance by 35% through query refactoring, distribution key optimization, and implementing result caching strategies.
- Built a robust unit testing framework using Jasmine, JUnit, and Mockito, achieving 70% code coverage across service, controller, and UI layers.
- Documented Software Design Specifications (SDS), requirements, validation documents, release notes, and branching strategies in Confluence for knowledge transfer.

Associate Software Developer

Nov 2020 - Aug 2021

Tata Consultancy Services (Client: FedEx)

Hyderabad, India

- Enhanced Employee Time Labor Management (ETLM) and User Provisioning modules using Java and Spring Boot, reducing manual provisioning errors by 40% through automated validation.
- Implemented Spring Boot CRON-based scheduled jobs for automated HR data synchronization, reducing bonus calculation processing time by 60% during peak cycles.
- Built a microservices architecture using Spring Cloud and Netflix Eureka for service discovery, integrating Apache Kafka messaging to reduce inter-service communication latency by 30%.
- Optimized database query performance by 25% through Spring Data JPA/Hibernate query tuning and strategic SQL Server indexing on high-traffic employee tables.
- Containerized Spring Boot microservices using Docker and orchestrated multi-environment deployments to Amazon ECS and EKS, streamlining development workflows.
- Integrated AWS SNS (Simple Notification Service) for real-time HR notifications, improving process reliability by 30%.
- Developed a real-time HR metrics dashboard using React with RxJS state management, providing business intelligence insights for workforce planning.
- Implemented automated end-to-end testing using Cypress, integrating tests into CI/CD pipelines (Jenkins, GitHub Actions) for consistent regression coverage.

Junior Developer

Dec 2018 - Nov 2020

Zoho Corporation

Chennai, India

- Developed modular UI components for JSP-based enterprise web applications following MVC architecture, ensuring code reusability across product lines.
- Implemented secure authentication using JWT tokens and OAuth 2.0 with Spring Security, enhancing system security and enabling SSO capabilities.
- Built a client-side validation framework using Angular Reactive Forms, improving data integrity and reducing backend errors by 20%.

- Refactored legacy Java code using Java 8 functional programming (Streams, Lambda expressions), improving code readability and performance by 15%.
- Optimized JVM performance through heap tuning, garbage collection optimization (G1GC), and thread pool configuration, reducing memory footprint by 35% and improving throughput by 25%.
- Designed a Logback configuration integrating Log4j with Spring Boot for structured logging and operational visibility.
- Optimized AWS infrastructure using EC2 Auto Scaling, S3 lifecycle management, and RDS Multi-AZ deployments, boosting availability by 40%.
- Configured AWS Application Load Balancer with health checks, improving application response time by 35%.
- Developed Python automation scripts for ETL workflows, reducing manual data processing time by 60%.
- Created GitHub Actions workflows using YAML for automated testing and multi-stage deployment pipelines.
- Integrated ESLint for code quality enforcement and identified/resolved 30+ critical bugs, enhancing system stability.

EDUCATION

Georgia State University

Master of Science in Computer Science | GPA: 3.63/4.0

Atlanta, GA

Aug 2021 – May 2023

Graduate Teaching Assistantship

- Delivered lectures for Operating Systems and Design Analysis of algorithms.
- Assisted students in the Java lab with code reviews, assignments, and technical queries.

Master's Thesis: Diabetic Retinopathy Severity Prediction

Developed a Python Flask application using VGG-16 and HANet deep learning architectures for automated diabetic retinopathy detection, achieving 85.27% accuracy on APTOS 2019 clinical dataset (20,000+ retinal images).

Vignan's Foundation for Science, Technology, and Research

Bachelor of Technology, Computer Science | GPA: 8.6/10.0

Guntur, India

Jun 2015 - Jul 2019